

# Yining Hua

Boston, MA

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LLM systems, RAG, benchmark design, agent evaluation, and high-stakes applied AI.

## Education

- **Harvard T.H. Chan School of Public Health**, Epidemiology (Ph.D.) Expected May 2028
- **Harvard Medical School**, Biomedical Informatics (MSc.) Sep 2021–Mar 2023
- **Smith College**, Computer Science (B.A.) Sep 2017–May 2021

## Technical

- **Programming & Tools:** Python, SQL, R, ChromaDB, Hex, Gemini, GPT-4o, Vercel
- **Focus Areas:** LLM systems, RAG, structured generation, agent evaluation, benchmark design, study design, causal inference, calibration and risk auditing

## Selected Research & Industry Experience

### Squirrel AI

Seattle, WA (Remote)

PhD Research Intern — LLM/RAG Systems

June 2025 – Mar 2026

- Designed an event-centric, graph memory-augmented agent (Python, ChromaDB, Gemini-2.5 on Vercel) with retrieve→re-rank→explain; cut inference cost by **96–98%**, reduced latency by **38%**, and improved relevant-memory recall by **24%**.
- Led and developed **SchemaRouter**, a hierarchical routing framework for JSON schema validation in LLM systems, achieving **1.5+** point gains over single-backend baselines through selective gate calibration and risk-coverage optimization.
- Designed an LLM evaluation system comparing single-response and multi-candidate generation under equal cost, showing capability gains up to **8.6 points** on GSM8K while quantifying safety/refusal tradeoffs, selector error versus oracle ceilings, and tail-risk with **VaR/CVaR**.

### Department of Digital Psychiatry, Beth Israel Deaconess Medical Center

Boston, MA

Researcher — LLM Evaluation

Nov 2024 – Aug 2025

- Co-built benchmark artifacts and reviewer playbooks used for continuous evaluation of mental-health agents.
- Developed evaluation frameworks and benchmark protocols for mental-health and general-health LLMs, with outputs published in **Nature Portfolio**, **World Psychiatry**, **JMIR AI**, and related venues.

### Department of Neurology, Massachusetts General Hospital

Boston, MA

PhD Researcher — ALS Prognostics, LLMs & Digital Twins

October 2025 – Present

- Develop evaluation and governance frameworks and tools for **AI/LLM development in high-stakes healthcare**, mapping model and study claims to minimal evidence packages.
- Develop **LLM-enabled prognosis models** for ALS and dementia using longitudinal EHR and trial data, combining sequence modeling, long-context clinical reasoning, and calibration auditing.
- Develop **digital twin** models integrating EHR and trial data to simulate patient trajectories and counterfactual outcomes; apply synthetic control arms and policy evaluation under real-world covariate shift.

### Dandelion Health

Boston, MA

PhD Intern — Core Analyst & Study Designer

July 2025 – Dec 2025

- Led a **\$650K** RWE collaboration with a top-10 pharma partner end-to-end; shipped protocol→cohort→analysis in **3 months**.
- Built reusable SQL+R pipelines for eligibility, anchors, follow-up, missingness, and causal models; standardized CRR endpoints across follow-on studies.

- Scaled GLP-1 target-trial emulation over **1.14M** echo records with **373,774** enrolled patients; reduced cohort build and run time from **35 hours to 0.4 hours** and launched Hex dashboards enabling clinician self-serve and faster sign-off.

## Harvard Medical School and Affiliated Hospitals

Boston, MA

Research Assistant — Clinical NLP/LLMs, Social Media Epi, Data Platforms

2021 – 2025

1. **Boston Children's Hospital (Urology)**: Developed LLM-based extraction pipelines for long clinical documents; compared active learning and clinician-in-the-loop review strategies using BioClinicalBERT, LoRA-adapted Llama 3.1 8B-Instruct, and GPT-4o, improving labeling accuracy by 6%.
2. **Brigham & Women's Hospital / Harvard Medical School**: Built reusable NLP and multimodal ML pipelines over large-scale social media and EHR data, including a harmonized **396M**-post Twitter corpus, Sentence-BERT topic modeling for misinformation studies, and structured-note models supporting **5+** downstream studies and **3** top-journal publications.

## Selected Publications

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1. **Hua Y**, Xia W., Bates D.W., Hartstein L., ..., Torous J. *Standardizing and scaffolding healthcare AI-chatbot evaluation*. **JMIR AI**. Accepted (2025).
2. Liu F, Li Z, Zhou H, ..., **Hua Y**, ..., Clifton D.A. *Large language models are poor clinical decision-makers: a comprehensive benchmark*. In: **EMNLP** (2024).
3. Liu J\*, Zhou P\*, **Hua Y\***, ... *Benchmarking Large Language Models on CMExam—A Comprehensive Chinese Medical Exam Dataset*. **NeurIPS** (2023).
4. Dwyer B., Flathers M., Sano A., ..., **Hua Y**, ... *MindBenchAI: an actionable platform to evaluate LLMs in mental healthcare*. **NPP Digit Psychiatry Neurosci**. Accepted (2025).
5. Ma J., Na H., Wang Z., **Hua Y**, Liu Y., Wang W., Chen L. *Detecting Conversational Mental Manipulation with Intent-Aware Prompting*. **COLING 2025**, pp. 9176-9183.
6. Ye Q., Liu J., Chong D., Zhou P., **Hua Y**, ... *Qilin-med: multi-stage knowledge injection medical LLM*. arXiv:2310.09089 (2023).

## Awards, Fellowships & Funding

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- **COLING 2025**: Best Short Paper.
- **NeurIPS 2022**: Spotlight (long paper); **NeurIPS 2023**: Travel Award.
- **IEEE ICHI 2023**: Best Poster; **AMIA 2022**: Distinguished Poster.